

► INTRODUCTION

Dementia affects more than 55 million people worldwide and is projected to double by 2050. Monitoring daily routine changes is essential for early detection of behavioral decline [1].

This project introduces an intelligent monitoring system capable of recognizing human activities such as mobility, sleep, personal hygiene, meals, and fall detection, while also identifying deviations from established routines. The system integrates data from a wrist-worn and an in-ear wearable device [2], enabling continuous, non-invasive, and real-time behavioral monitoring in elderly individuals with dementia.

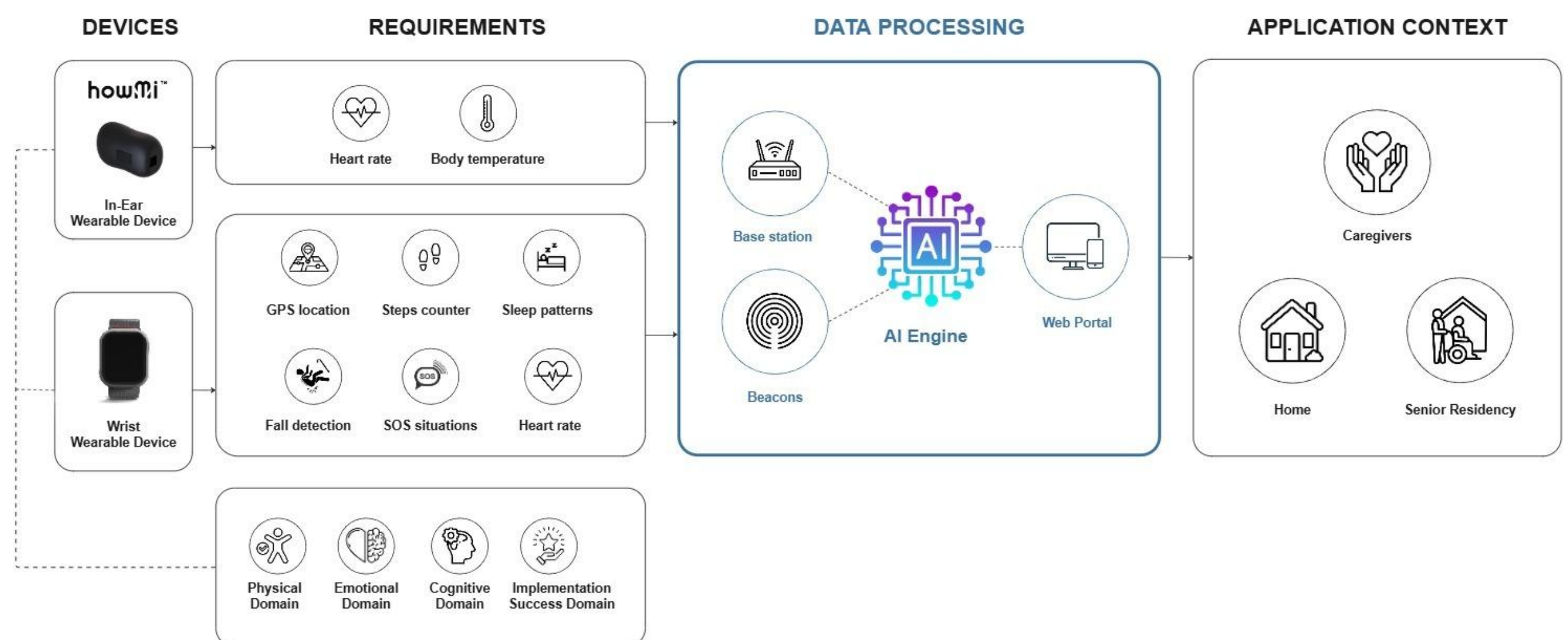


Figure 1. System architecture wearable wrist and ear devices, a local AI engine, and a web-based caregiver portal.

► METHODOLOGY

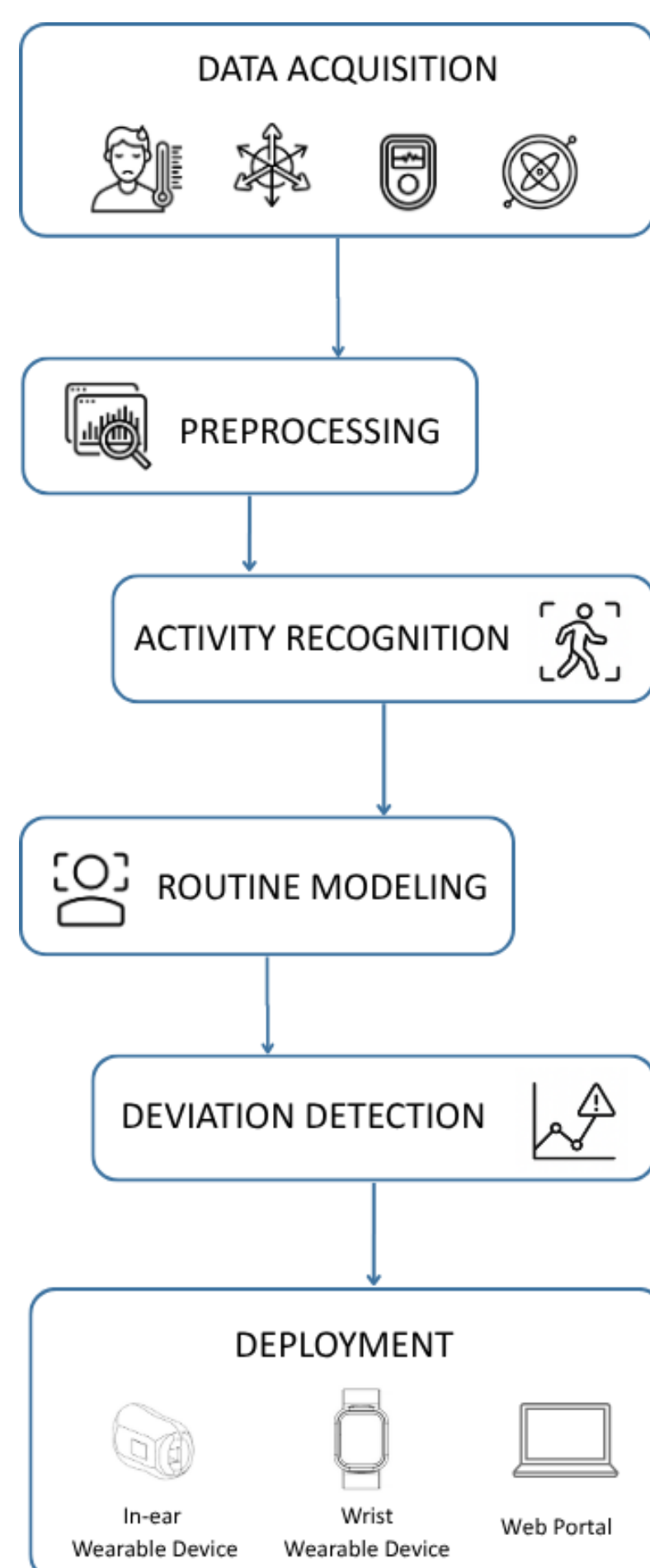


Figure 2. AI pipeline

► OBJECTIVES

The main objective of this work is to develop an AI-based wearable system for monitoring daily activities and routines of elderly people with dementia.

Specific Objectives:

- Study existing datasets.
- Develop and evaluate machine learning algorithms for activity recognition.
- Collect and preprocess sensor data.
- Integrate real-time monitoring algorithms for routine analysis.

► RESULTS AND FUTURE WORK

Baseline models for activity classification have been implemented following the completion of data preprocessing. These models are being evaluated using standard classification metrics, including precision, recall, and F1-score. A key challenge identified during this stage is the high variability across individuals and the limited availability of real data from people with dementia, which poses constraints to model generalizability and validation [3].

Data collection from healthy volunteers simulating underrepresented daily activities, such as hygiene routines, will take place to support further model development. The next phase will involve the simulation of behavioral anomalies to validate the system's capacity for anomaly detection in realistic conditions. The system will then undergo tests focused on accuracy, robustness, and adaptability to ensure reliable performance in real-world use by elderly individuals with dementia.

The expected outcome is a robust and adaptable model that can be integrated into a wearable prototype, supporting continuous, autonomous, and personalized monitoring, while providing timely and actionable insights for caregivers.

► ACKNOWLEDGEMENTS

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► BIBLIOGRAPHY

- [1] Rocha, I. C., Arantes, M., Carvalho, M., Moreira, A., Vilaça, J. L., Morais, P., Matos, D., & Carvalho, V. (2024). *Development of a wearable device for monitoring the activity of elderly people with dementia: first insights*. <https://doi.org/10.54499/UIDB/05549/2020>
- [2] Rocha, I. C., Arantes, M., Moreira, A., Vilaça, J. L., Morais, P., Matos, D., & Carvalho, V. (2024). Monitoring Wearable Devices for Elderly People with Dementia: A Review. In *Designs* (Vol. 8, Issue 4). Multidisciplinary Digital Publishing Institute (MDPI). <https://doi.org/10.3390/designs8040075>
- [3] Carvalho, M., Rocha, I., Arantes, M., Linhares, R., Soares, J., Moreira, A., Vilaça, J., Matos, D., Morais, P., & Carvalho, V. (2025). Powered Wearable Technologies for Dementia Care: Evaluating Activity Recognition Models and Dataset Challenges. *Proceedings of the 18th International Joint Conference on Biomedical Engineering Systems and Technologies*, 995–1006. <https://doi.org/10.5220/0013396600003911>